

100 shares.⁶⁷ Market participants interested in quotation data for orders that have a size less than a round lot, i.e., odd-lots, must purchase individual exchange proprietary feeds.⁶⁸ This odd-lot order information is highly relevant to market participants, including for investors who trade small numbers of shares.

The MDI Rules expanded the NMS information that will be made available for dissemination within the national market system in order to increase transparency about better prices available in the market.⁶⁹ The Commission, in the MDI Rules, amended Regulation NMS to include a definition of “round lot” that assigns each NMS stock to a round lot size based on the stock’s average closing price. The round lot definition, once implemented, will increase transparency about smaller sized orders in higher priced stocks by assigning NMS stocks priced over \$250 to round lot sizes that are less than the predominant 100 shares.⁷⁰ The Commission also adopted a definition of odd-lot information as part of the MDI Rules.⁷¹ Once implemented, information regarding the prices and sizes of odd-lot orders priced better than the NBBO will be

⁶⁷ According to NYSE Trade and Quote (“TAQ”) Data, as of Nov. 28, 2023, 11 NMS stocks have a round lot size other than 100. Nine NMS stocks have a round lot size of 10 and two NMS stocks have a round lot size of one share.

⁶⁸ See Proposing Release, supra note 11, at 80294; MDI Adopting Release, supra note 10, at 18599.

⁶⁹ See Proposing Release, supra note 11, at 80270.

⁷⁰ 17 CFR 242.600(b)(93). In the MDI Adopting Release, the Commission stated that “[d]efining smaller-sized orders in higher-priced stocks as round lots, in addition to providing transparency into such quotations, ensures that these smaller-sized orders can establish the [national best bid and national best offer], receive order protection, and invoke the applicability of several other rules under Regulation NMS.” See MDI Adopting Release, supra note 10, at 18613.

⁷¹ Preexisting 17 CFR 242.600(b)(69). “Odd-lot information” is defined as (1) odd-lot transactions, and (2) odd-lots at a price greater than or equal to the national best bid and less than or equal to the national best offer, aggregated at each price level at each national securities exchange and national securities association. Id.

made available within the national market system and is expected to be made widely available to investors.⁷²

For the reasons explained in the MDI Adopting Release, the MDI Rules sequenced the implementation of these definitions in the later stages of the implementation schedule.⁷³ The implementation of the MDI Rules began with the filing of amendments to the effective national market system plan(s) as required under Rule 614(e) (“MDI Plan Amendments”).⁷⁴ The Operating Committees of the CTA/CQ Plan and UTP Plan⁷⁵ filed the proposed MDI Plan Amendments on November 5, 2021,⁷⁶ which the Commission disapproved.⁷⁷ As a result, the

⁷² The Commission stated that the inclusion of this odd-lot quotation information would allow market participants “to trade in a more informed and effective manner,” and that “the new definition of round lot and the increased availability of better priced odd-lot information will provide investors with valuable information about the best prices available and help to facilitate more informed order routing decisions and the best execution of investor orders.” See MDI Adopting Release, supra note 10, at 18602 and 18613. Unlike orders in the round lot sizes adopted pursuant to the MDI Rules, odd-lots are not “protected quotations.” See 17 CFR 242.600(b)(16), (81), (82).

⁷³ See MDI Adopting Release, supra note 10, at 18698. Pursuant to the implementation schedule of the MDI Rules, the round lot definition was set to be implemented as part of the last phase and odd-lot quotation information was set to be implemented during a “parallel operation period.” See id. at 18700-01. As originally adopted, during the parallel operation period, the exclusive SIPs would have continued to disseminate the data that they currently disseminate and competing consolidators would have been permitted to offer consolidated market data products, including odd-lot information. Because the round lot definition would have been implemented during a later phase, the exclusive SIPs and competing consolidators would have collected, consolidated and disseminated NMS information based on then current exchange definitions of round lot. Id. at 18699-18701.

⁷⁴ 17 CFR 242.614(e). The Commission’s approval of amendments to the effective national market system plan(s) filed pursuant to rule 614(e) will be the starting point for the rest of the MDI Rules implementation schedule, which includes a 180-day development period, during which competing consolidators can register with the Commission, and ends with the cessation of the operations of the exclusive SIPs and testing and implementation of the changes necessary to implement the round lot definition. See MDI Adopting Release, supra note 10, at 18699-701; Proposing Release, supra note 11, at 80295.

⁷⁵ See supra note 60.

⁷⁶ See Securities Exchange Act Release Nos. 93615 (Nov. 19, 2021), 86 FR 67800 (Nov. 29, 2021); 93625 (Nov. 19, 2021), 86 FR 67517 (Nov. 26, 2021); 93620 (Nov. 19, 2021), 86 FR 67541 (Nov. 26, 2021); 93618 (Nov. 19, 2021), 86 FR 67562 (Nov. 26, 2021).

⁷⁷ See Securities Exchange Act Release Nos. 95848 (Sept. 21, 2022), 87 FR 58544 (Sept. 27, 2022); 95849 (Sept. 21, 2022), 87 FR 58592 (Sept. 27, 2022); 95850 (Sept. 21, 2022), 87 FR 58560 (Sept. 27, 2022); 95851 (Sept. 21, 2022), 87 FR 58613 (Sept. 27, 2022).

participants to the effective national market system plan(s) will need to develop and file new proposed amendments pursuant to Rule 608.⁷⁸

2. Proposed and Adopted Amendments

In light of the delays in the implementation of the MDI Rules, the Commission proposed to accelerate the implementation of the round lot and odd-lot information definitions, to allow investors to benefit sooner from greater transparency and accessibility of better priced orders and improved execution quality.⁷⁹ As discussed further below, the Commission is accelerating the implementation of the round lot and odd-lot information definitions but is providing the industry with more time to make the necessary systems changes to implement these definitions than what was proposed.⁸⁰

Additionally, the Commission proposed to amend the definition of odd-lot information to include a new data element for the best odd-lot orders available in the market, which would be

⁷⁸ On Sept. 1, 2023, the Commission ordered the exchanges and the Financial Industry Regulatory Authority, Inc. (“FINRA”) to file a new single national market system plan regarding consolidated equity market data. See Securities Exchange Act Release No. 98271, 88 FR 61630 (Sept. 7, 2023). On Jan. 19, 2024, the Commission published notice of filing of a National Market System Plan for Consolidated Equity Market Data. See Securities Exchange Act Release No. 99403, 89 FR 5002 (Jan. 25, 2024). On April 23, 2024, the Commission instituted proceedings pursuant to Rule 608(b)(2)(i) of Regulation NMS to determine whether to approve or disapprove the proposed plan or to approve the proposed plan with any changes or subject to any conditions the Commission deems necessary or appropriate after considering public comment. See Securities Exchange Act Release No. 100017, 89 FR 33412 (Apr. 29, 2024). On July 11, 2024, the Commission extended the period within which to conclude proceedings regarding the proposed plan to 240 days from the date of publication of the notice. See Securities Exchange Act Release No. 100500 (Jul. 11, 2024), 89 FR 58235 (Jul. 17, 2024).

⁷⁹ See Proposing Release, supra note 11, at 80299; see also infra sections V.B.2. and V.C.1. In addition, as discussed below, the Commission is amending the definition of round lot so that the frequency of round lot changes will be consistent with the frequency of minimum pricing increment changes under amended Rule 612. See infra section V.B.3.b. The Commission is not changing the calculation used to assign round lots or the round lot tiers in the round lot definition adopted in the MDI Rules.

⁸⁰ See infra sections V.B.3, V.C.1, and VI.C.

made available to investors broadly. The Commission is adopting the best odd-lot order to buy and the best odd-lot order to sell as part of odd-lot information as proposed.⁸¹

D. Overarching Comments on the Proposing Release

The Commission received comments from a variety of market participants on the Proposing Release.⁸²

Many commenters broadly supported the Regulation NMS Proposal.⁸³ Two commenters urged the Commission to promptly adopt the Regulation NMS Proposal.⁸⁴ One commenter urged the Commission to revise and adopt the Rule 605 Proposal as well as the Regulation NMS

⁸¹ See infra section V.C.2.

⁸² The comment letters on the Proposing Release (File No. 7-30-33) are available at <https://www.sec.gov/comments/s7-30-22/s73022.htm>.

⁸³ See, e.g., Letters from Mark Rogers dated Mar. 30, 2023 (“I approve of the proposed changes to Regulation NMS”); Omar Fakhro dated Mar. 28, 2023 (“I as a household investor strongly support this rule for a better and fair market for EVERYONE”); Danielle Ball dated Mar. 27, 2023 (“The proposed tick size regime, variable minimum pricing increment model, and revised round lot definition are important steps towards promoting fair and transparent pricing across trading venues.”); Keith Noble dated Apr. 1, 2023; Chris Miller dated Apr. 1, 2023; Kristen Palmer dated Apr. 1, 2023; Amanda Kappes dated Apr. 1, 2023; Ian Rohel, dated Apr. 1, 2023; Riley Hume dated Apr. 1, 2023; Matt Kelleher dated Apr. 1, 2023; Keagan Wethington dated Mar. 31, 2023; J. W. Verret, Associate Professor, George Mason University Antonin Scalia Law School, dated Jan. 12, 2024 (“Verret Letter III”) at 26 (“...the proposed amendments to Reg NMS rules regarding minimum pricing increments and the proposed reforms to volume/access fees both support the core principles of free market economics and will lead to a more competitive, transparent, and efficient market landscape.”); Eric Budish, Paul G. McDermott Professor of Economics and Entrepreneurship, The University of Chicago Booth School of Business, dated Jan. 18, 2024 (“Budish Letter”) at 1 (“... this set of rules changes – primarily, a finer tick-size for tick-constrained stocks, a lower access fee cap, and harmonization of pricing increments for on-exchange and off-exchange trading – will reduce both investors’ costs and the overall complexity of U.S. equity markets.”); Stephen W. Hall, Legal Director and Securities Specialist, and Brady Williams, Legal Counsel, Better Markets, Inc., dated Mar. 31, 2023 (“Better Markets Letter I”) at 8-17; Joseph Saluzzi, Partner, Themis Trading LLC, dated Mar. 31, 2023 (“Themis Letter”) at 2-8; John Ramsay, Chief Market Policy Officer, Investors Exchange LLC, dated Mar. 20, 2023 (“IEX Letter I”); Letter Type A, of which 22 comments were received; Letter Type C, of which 5 comments were received; Letter Type D, of which 255 comments were received; Letter Type E, of which 14 comments were received; Letter Type G, of which 652 comments were received; Letter Type H, of which 853 comments were received; Letter Type I, of which 22 comments were received; Letter Type J, of which 15 comments were received; Letter Type K, of which 22 comments were received; and Letter Type L, of which 4 comments were received; available at <https://www.sec.gov/comments/s7-30-22/s73022.htm>.

⁸⁴ See, e.g., Letters from Tyler Gellasch, President & CEO, Healthy Markets Association, dated Mar. 31, 2023 (“Healthy Markets Letter I”) at 28, 31; J. W. Verret, Associate Professor, George Mason University Antonin Scalia Law School, dated Sept. 20, 2023 (“Verret Letter I”) at 1-2, 4, 5.

Proposal without delay.⁸⁵ Another commenter suggested that the Commission prioritize the adoption of the Regulation NMS Proposal⁸⁶ stating that, of the four EMS Proposals related to equity market structure, the Regulation NMS Proposal “is the least controversial and the least interdependent on the other three, and so is the easiest one for the Commission to move forward”⁸⁷ and “has garnered the most consensus and support from various market participants.”⁸⁸

Some commenters agreed that Rules 610 and 612 should be amended but recommended that the proposed amendments be modified and that the Commission consider more modest, incremental changes to minimize the possibility of unintended consequences and to enable the Commission and market participants to evaluate the impact of the changes on trading and execution quality.⁸⁹

The issues related to the amended rules have been considered by the Commission and market participants for several years.⁹⁰ Further, the Commission has analyzed data provided by market participants and conducted its own data analysis to inform the amendments that were included in the Proposing Release and in this release.⁹¹ The Commission has evaluated the

⁸⁵ See Healthy Markets Letter I at 28, 31.

⁸⁶ See Verret Letter I at 1.

⁸⁷ Verret Letter I at 1-2.

⁸⁸ See Verret Letter I at 2 (stating that the Regulation NMS Proposal is supported by “a wealth of prior work by the Commission in the form of a pilot tick size study, comments submitted to the SEC regarding the transaction fee pilot, and numerous roundtables and proceedings of the SEC’s Investor Advisory Committee and SEC’s Equity Market Structure Advisory Committee.”).

⁸⁹ See, e.g., *infra* note 92.

⁹⁰ See, e.g., EMSAC Archives, *supra* note 4 (Rule 610 was considered at the EMSAC), see also *supra* note 4 (discussing the EMSAC); *infra* note 362 and accompanying text for a discussion of previous considerations of Rule 610. For a discussion of previous considerations of Rule 612, see Proposing Release, *supra* note 11, at 80272.

⁹¹ See *infra* sections V.B.1; V.B.3.b.iv and VII.D.

national market system and its operation in light of changes in the market and has sought input from market participants throughout this process.⁹² After considering the comments, which are discussed in context below, the Commission is adopting amendments to these rules with certain modifications from the Proposing Release.

The Commission received several comments that addressed the interaction between the different individual proposed rule amendments that made up the Regulation NMS Proposal. One commenter stated that adopting the proposed changes to the minimum pricing increments in proposed Rule 612 along with the proposed acceleration of the round lot definition and the proposed access fee caps in Rule 610 “would impact the value of providing liquidity on public markets and consequently would raise costs for investors,” and urged the Commission to review how these changes would together impact liquidity.⁹³ The Commission has considered the impact of the amendments on liquidity and does not believe that they will raise costs for investors.⁹⁴ On the contrary, as discussed further below, the amendments will enhance the ability of market participants to price their orders in a competitive manner, reduce the amount of fees for accessing protected quotations, help to ensure that exchange fees are knowable when an order

⁹² See also Proposing Release, *supra* note 11, at 80272 (discussing considerations of minimum pricing increments since Rule 612 was adopted) and 80287 (discussing considerations of access fee caps since Rule 610 was adopted). See also IEX Letter I at 5 (describing steps taken by the Commission since the adoption of Regulation NMS in 2005 to review the impact of Regulation NMS, including the solicitation of input from stakeholders, further stating, “[t]he history shows that the Commission’s current Proposals do not arise in a vacuum. In fact, the Commission has deliberately considered the views of multiple stakeholders over years of review, and its current Proposals grow out of and build on that ongoing review.”).

⁹³ See Letter from Naureen Hassan, President, UBS Americas, Robert Karofsky, President, UBS Investment Bank, and Suni Harford, President, UBS Asset Management, dated Mar. 31, 2023 (“UBS Letter”) at 10. See *infra* section V.B.3.b.i and section VII.D.4.a for discussions of the interaction between the round lot definition and the proposed changes to the minimum pricing increments.

⁹⁴ See *infra* section VII.D.4.a (explaining that the interaction of the reduction in tick size and the MDI Rules’ round lot definition would not have a material impact on the NBBO for affected stocks as such stocks would be exceptionally liquid, which should protect their NBBO from material deterioration).

is placed and provide transparency about orders in the market that are priced better than the NBBO. These changes will enhance the operation of the national market system and provide significant benefits to investors.

Another commenter stated that the Regulation NMS Proposal would increase “market data costs” because retail brokers would have to take in and store an increased amount of market data to comply with the changing minimum pricing increments, the MDI Rules’ round lot definition, and the odd-lot information requirements and to update their systems accordingly, and because the exclusive SIPs may cause third-party data vendors to require additional hardware to support higher message rates.⁹⁵ As discussed below,⁹⁶ the Commission is adopting amendments to the minimum pricing increments with modifications from the proposal, which will lessen the potential costs identified by the commenter. Specifically, the Commission is adopting one minimum pricing increment for a smaller universe of NMS stocks than was proposed and is reducing the frequency of minimum pricing increment updates from a quarterly to a semiannual basis.⁹⁷ While this additional minimum pricing increment will likely require market participants to incur new technology costs to manage the new data, fewer changes are being adopted than were proposed and these changes are necessary and justified to address the issues related to constraints that have developed with the \$0.01 minimum pricing increment.⁹⁸ Further, the costs

⁹⁵ See Letter from Derrick Chan, Head of Equities, Fidelity Capital Markets, dated Mar. 31, 2023 (“Fidelity Letter”) at 17. The commenter described “market data costs” as those related to systems changes necessary to implement the new minimum pricing increments, round lot definition, and odd-lot information definition.

⁹⁶ See infra section III.C.

⁹⁷ See infra section III.C.7.a; section III.C.8; section VII.D.1.d section and section VII.F.1.c

⁹⁸ See infra section VII.A; section VII.D.1.c (responding to comments raising concerns about increased message traffic increasing costs and stating: “[t]he Commission recognizes the potential for these costs articulated by the commenters but, considering additional information provided by commenters, expects these effects to be mild – including the effect on CAT costs.”).

related to implementing the round lot definition were considered as part of the MDI Rules and the acceleration of the timing of implementation does not increase those costs. Although the Commission is modifying the round lot definition from the definition adopted in the MDI Rules, the modifications will reduce ongoing round lot implementation costs because round lots will be assigned less frequently, i.e., from a monthly basis to a semiannual basis, which means that systems will have to be updated less frequently. Synchronizing the dates of the changes to round lots and minimum pricing increments should also lower ongoing implementation costs for market participants by potentially decreasing the number of updates needed for their trading systems.⁹⁹ Finally, the costs related to implementing the odd-lot information definition were considered in the Proposing Release.¹⁰⁰ The adopted amendments, which will result in fewer systems changes than anticipated in the Proposing Release, will result in lower implementation costs than were contemplated in the proposal¹⁰¹ and reduce the amount of data disseminated by the exclusive SIPs and any future competing consolidators as compared to what was contemplated in the Proposing Release.

One commenter stated that the implementation of various components of the Proposing Release at or around the same time (specifically access fees, minimum pricing increments and round lot sizes) could complicate the Commission’s ability to assess the impact of a specific change and “whether other consequences will ensue.”¹⁰² To the specific concerns of this

⁹⁹ See infra notes 1594-1595 and accompanying text.

¹⁰⁰ See Proposing Release, supra note 11, at 80334.

¹⁰¹ See infra section VII.D.5.

¹⁰² See Letter from Rich Steiner, Head of Global Market Structure, RBC Capital Markets, dated Mar. 31, 2023 (“RBC Letter”) at 2. See also Letter from Nathaniel N. Evarts, Managing Director, Head of Trading, Americas, and Kimberly Russell, Market Structure Specialist, Global SPDR Business, State Street Global Advisors, dated Mar. 30, 2023 (“State Street Letter”) at 5 (suggesting that the amendments to reduce the

commenter, the Commission has carefully considered the interacting effects of access fees, minimum pricing increments, and round lot sizes, see section VII. While the Commission acknowledges that staging amendments may make them easier to study, the nature of the adopted amendments will still make such study possible, even if implemented together. Namely, the set of stocks for which the tick size change applies tends to differ from the set of stocks for which round lot changes apply.¹⁰³ Access fee changes apply to some stocks that will not be directly affected by either round lot reform or tick size changes. Further, staging the amendments would delay the significant benefits of the amendments.¹⁰⁴

Several commenters suggested implementing the proposed accelerated implementation of the round lot and odd-lot information definitions so that the effects of these definitions could inform other proposed changes.¹⁰⁵ Other commenters suggested that round lots should be implemented before the proposed changes to the minimum pricing increments, so that data based on the MDI Rules' round lots could inform changes to the minimum pricing increments.¹⁰⁶

access fee caps should be implemented before the minimum pricing increments to isolate the impact of the effects) and infra section VII.D.2.c (responding to the State Street Letter).

¹⁰³ See infra note 801 for analysis identifying only two stocks that would have qualified for both the tick reduction and a reduction in the round lot as of Nov. 30, 2023. See also infra section VII.D.4.a for a discussion of the small overlap of the round lot definition and the tick size change.

¹⁰⁴ See, infra, section II. The Commission recognizes that delaying the rule would likewise delay costs to affected parties.

¹⁰⁵ See, e.g., Letters from Jennifer W. Han, Executive Vice President, Chief Counsel & Head of Global Regulatory Affairs, Managed Funds Association, dated Mar. 30, 2023 ("MFA Letter") at 14; Sarah A. Bessin, Deputy General Counsel, and Nhan Nguyen, Assistant General Counsel, Investment Company Institute, dated Mar. 31, 2023 ("ICI Letter I") at 2, 7; Gerald O'Reilly, Co-CEO and Chief Investment Officer, and Ryan Wiley, Global Head of Equity Trading, Dimensional Fund Advisors LP, dated Mar. 31, 2023 ("Dimensional Letter") at 2.

¹⁰⁶ See Letter from Hubert De Jesus, Managing Director, Global Head of Market Structure and Electronic Trading, and Samantha DeZur, Managing Director, Global Public Policy Group, BlackRock, Inc., dated Mar. 31, 2023 ("BlackRock Letter") at 17; Dimensional Letter at 2.

While the dissemination of odd-lot information will result in the display of narrower spreads based on odd-lots, the calculation of the TWAQS for determining minimum pricing increments is based on round lots.¹⁰⁷ Therefore, odd-lot information will not have an impact on determining minimum pricing increments under Rule 612. Further, for the reasons discussed below, the interaction of the reduction in tick size and the MDI Rules' round lot definition will likely not have a material impact on the NBBO of affected stocks since only the most exceptionally liquid stocks would have prices over \$250 and a TWAQS equal to or less than \$0.015.¹⁰⁸ Therefore, it is not necessary to postpone amending the minimum pricing increments until data is analyzed using the MDI Rules' round lots.

In addition, the dissemination of odd-lot information in conjunction with the MDI Rules' round lot sizes will increase transparency about better priced orders and therefore should be implemented within a similar time frame.¹⁰⁹ Odd-lot information will be provided for all NMS stocks, not just those NMS stocks that may be assigned a smaller round lot. As discussed below, the number of NMS stocks that may be assigned a smaller round lot as of November 30, 2023 is 163 NMS stocks.¹¹⁰ Therefore, while the MDI Rules' round lot sizes will provide transparency about some better priced orders in higher priced stocks, they will not enhance transparency about those orders that continue to be defined as odd-lots and will not increase transparency for NMS stocks priced at \$250 or less. This transparency is important for investors as it will enhance their ability to assess the current pricing in the market for certain NMS stocks. Therefore, the odd-lot

¹⁰⁷ See infra section III.C.7.b.

¹⁰⁸ See infra section V.B.3.b.i (identifying only two stocks – both highly liquid – that would have qualified for both a tick reduction and a reduction in the round lot as of Nov. 30, 2023).

¹⁰⁹ See infra section VII.D.4.

¹¹⁰ Id.